

Group 1 Proposal

The Travelling Salesman Problem

One of the most famous search problems in the world by far is the traveling salesman problem. But why is it so famous? Its prominence is majorly around how simple it seems at first glance, being what's the minimum distance the salesman can take to deliver all their goods to the correct points. However, complexity arises when there are many different points the salesman has to take, in which case the minimum distance is not always obvious and typically involves what appear to be detrimental paths, but in the end gives the better distance. Group one found this topic fascinating and possibly even helpful in the future, as "solving" the traveling salesman problem opens the door to similar real-life cases, for example, scheduling to use your time most efficiently. To find a sufficient solution to the traveling salesman problem, it is necessary to compare multiple search algorithms/methods to determine which can best find the minimum distance. While multiple search algorithms exist with major differences, such as depth-first search and breadth-first search, these are obvious changes we can make; some smaller changes that can greatly enhance or hurt an algorithm include adding a greedy algorithm/path initially, the cost function used, etc. One thing group one needs to keep in mind to ensure fairness between the algorithms is to have a variety of destinations in terms of number and distance; otherwise, the chosen search algorithm could overlook faults, such as only finding the minimum distance when the number of destinations is low. Overall, group one will test multiple search methods with varying intricacies across a variety of differently formed traveling salesman problems to determine which method finds the shortest distance needed to travel.

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